Formula Student spaceframe: torsional stiffness sensitivity by tube group

Baseline 1712 Nm/deg at 34.18 kg (50.1 Nm/deg/kg). Load case: 408.6 Nm applied at the front axle pickups, all three translations fixed at the four rear pickups. Every row is a rules-legal frame from the T3 catalogue.

Tube group	Current	Mass (kg)	Strain energy	Free upgrade (no extra mass)	ΔNm/deg	Best value upgrade	+g	ΔNm/deg	Nm/deg per g	Rank
MAIN_HOOP	25x2.5	3.79	14.0%	30x2	+100.4	26.9x2.3	22	+42.7	1.923	1
PICKUP_SUPPORT	25.4x1.6	1.81	6.6%	41.3x0.9	+56.8	44.5x0.9	55	+80.3	1.452	2
HARNESS_BAR	25x2.5	0.62	5.1%	30x2	+36.1	31.75x2	36	+50.5	1.400	3
FRONT_HOOP	25x2.5	2.25	5.1%	30x2	+35.1	26.9x2.3	13	+14.7	1.116	4
FH_BRACE	25.4x1.6	2.88	6.5%	26.9x1.5	+13.3	33.7x1.2	69	+66.5	0.957	5
MH_BRACE	25.4x1.6	2.13	3.6%	26.9x1.5	+6.4	33.7x1.2	52	+38.1	0.739	6
FBH	25.4x1.6	2.25	3.9%	26.9x1.5	+8.5	33.7x1.2	59	+43.2	0.733	7
REAR_IMPACT	25.4x1.6	2.48	18.4%	26.9x1.5	+5.5	33.7x1.2	60	+35.3	0.591	8
HARNESS_SUPPORT	25.4x1.6	1.45	1.5%	26.9x1.5	+3.5	33.7x1.2	35	+20.1	0.576	9
REAR_BAY	25.4x1.2	3.58	3.2%	31.75x0.9	+20.5	33.7x0.9	59	+30.3	0.513	10
FLOOR	25.4x1.2	2.00	1.7%	31.75x0.9	+10.4	33.7x0.9	33	+16.3	0.493	11
SIS_LOWER	25.4x1.6	1.22	6.5%	26.9x1.5	+1.5	33.7x1.2	30	+10.9	0.371	12
SIS_DIAGONAL	25.4x1.6	1.38	6.1%	26.9x1.5	+0.9	33.7x1.2	33	+7.3	0.218	13
SIS_UPPER	25.4x1.6	1.19	0.5%	26.9x1.5	+0.9	33.7x1.2	29	+5.4	0.187	14
FBS_UPPER	25.4x1.2	1.03	4.3%	-	-	35x1.2	410	+56.5	0.138	15
FBS_LOWER	25.4x1.2	0.98	3.7%	-	-	26.9x1.2	61	+7.5	0.124	16
MHB_SUPPORT	25.4x1.2	2.05	8.3%	-	-	26.9x1.2	127	+10.3	0.082	17
FBS_DIAGONAL	25.4x1.2	1.09	1.1%	-	-	44.5x1.2	857	+30.3	0.035	18

How to read this. *Strain energy* is the share of the frame's elastic energy held by that group under torsion: a group holding 5% cannot buy more than 5%, however much steel goes into it. *Free upgrade* is a different catalogue tube of the same or lower mass that is stiffer, legal because Table 4 sets minimum wall, area and second moment but not a diameter, and I grows with OD^4 while mass grows with area. *Best value upgrade* is the largest Nm/deg per gram added. Changes are measured one group at a time against the baseline, so they are not additive: groups share load paths, and combining them is what run_optimise.py does.

Method. 3D Euler-Bernoulli beam elements, 6 DOF per node, written from scratch in NumPy. Validated against closed-form cantilever solutions (~1e-10) and cross-checked against PyNite, a third-party solver (~1e-12). Buckling by Euler above the transition slenderness and the Johnson parabola below it, against welded yield strength. Geometry is rules-informed and verified against section T3; cockpit templates, Percy clearance, roll hoop bend geometry and SES approval are out of scope, so this is not a scrutineering result.